MULTIVARIATE OPTIMIZATION WITH EQUALITY CONSTRAINTS

Fundamentals of optimization

Multivariate optimization with constraints

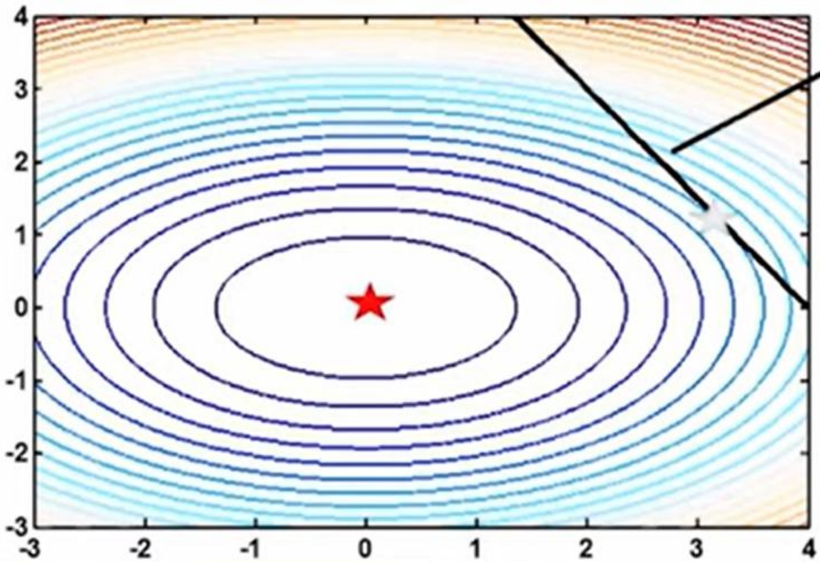
$$\min_{x_1, x_2} \; 2x_1^2 + 4x_2^2$$

$$st$$

$$3x_1 + 2x_2 = 12$$

Constrained minimum

Unconstrained minimum



$3x_1 + 2x_2 = 12$

All points on this line represent the feasible region

Unconstrained minimum is not the same as constrained minimum

Fundamentals of optimization

$f(x)$

$f(x, \lambda, \ldots,$

Multivariate optimization with equality constraints

At optimum (one equality constraint case)

$$-\nabla f(\overline{x^*}) = \lambda^* \nabla h(\overline{x^*})$$

In higher dimensions and when there are more than one equality constraint

$$-\nabla f(\overline{x^*}) = \sum_{i=1}^{l} [\nabla h_i(\overline{x^*})] \lambda_i^*$$

Gradient lies in the space spanned by the normal of the gradients

Fundamentals of optimization

Multivariate optimization with equality constraints

$f(x)$

$f(x_1, x_2, \ldots, x_n)$

Min $f(x_1, x_2, \ldots x_n)$

$h(x_1, x_2, \ldots x_n) = 0$

At optimum (one equality constraint case)

$$-\nabla f(\overline{x^*}) = \lambda^* \nabla h(\overline{x^*})$$

$-\nabla f = \lambda \nabla h$

In higher dimensions and when there are more than one equality constraint

$$-\nabla f(\overline{x^*}) = \sum_{i=1}^{l} [\nabla h_i(\overline{x^*})]\, \lambda_i^*$$

Gradient lies in the space spanned by the normal of the gradients

$-\nabla f = \lambda \nabla h$

$f(x)$

$h_1(x) = 0$

$h_2(x) = 0$

$-\nabla f = \lambda_1 \nabla h_1 + \lambda_2 \nabla h_2$

$h_1(x) = 0;\ h_2(x) = 0$

Fundamentals of optimization

Multivariate optimization

$$\min_{x_1, x_2} \; 2x_1^2 + 4x_2^2$$

$$st$$

$$3x_1 + 2x_2 - 12 = 0$$

$$-\nabla f = \lambda \nabla h$$

$$h(x) = 0$$

$$-\nabla f = -\begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{pmatrix} = \begin{pmatrix} -4x_1 \\ -8x_2 \end{pmatrix}$$

$$\begin{pmatrix} \frac{\partial h}{\partial x_1} \\ \frac{\partial h}{\partial x_2} \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

First order condition

$$\begin{aligned} -4x_1 &= 3\lambda \\ -8x_2 &= 2\lambda \\ (3x_1 + 2x_2 - 12) &= 0 \end{aligned} \quad \xrightarrow{\text{solving}} \quad \begin{bmatrix} x_1^* \\ x_2^* \\ \lambda^* \end{bmatrix} = \begin{bmatrix} 3.27 \\ 1.09 \\ -4.36 \end{bmatrix}$$

MULTIVARIATE OPTIMIZATION WITH INEQUALITY CONSTRAINTS

General formulation

Multivariate optimization

$$\min_{\bar{x}} \quad f(\bar{x})$$

st

$$h_i(\bar{x}) = \bar{0}, \; i = 1, \dots m$$

$$g_j(\bar{x}) \leq \bar{0}, \; j = 1, 2 \dots l$$

Necessary condition for $\overline{x^*}$ to be the minimizer

KKT conditions has to be satisfied

Sufficient condition

$\nabla^2 L(\overline{x^*})$ has to be positive definite

Summary – KKT conditions

Multivariate optimization

When both equality and inequality constraints are present, at the optimum we have

KKT (Karush-Kuhn-Tucker) conditions

$$\nabla f(\overline{x^*}) + \sum_{i=1}^{l} [\nabla h_i(\overline{x^*})]\ \lambda_i^* + \sum_{j=1}^{m} [\nabla g_j(\overline{x^*})] \mu_j^* = 0$$

Gradient of the "Lagrangian function" at x^*

$$L(\overline{x^*}, \lambda^*, \mu^*) = f(\overline{x^*}) + \sum_{i=1}^{l} \lambda_i h(\overline{x^*}) + \sum_{j=1}^{m} \mu_j g_j(\overline{x^*})$$

$$h_i(x^*) = 0, i = 1 \dots l$$

$$\lambda_i \epsilon R, i = 1 \dots l$$

Ensures that the optimum satisfies equality constraints

$$g_j(\overline{x^*}) \leq 0, j = 1 \dots m$$

Ensures that the optimum is in the feasible region

$$\mu_j^* (g_j(\overline{x^*}) = 0$$

Complementary slackness

$$\mu_j^* \geq 0, j = 1 \dots m$$

- No possibility of improvement near the active constraints

Summary – KKT conditions

Multivariate optimization

- In general it is difficult to use the KKT conditions to solve for the optimum of an inequality constrained problem (than for a problem with equality constraints only) because we do not know *a priori* which constraints are active at the optimum.

- Makes this a combinatorial problem

- KKT conditions are used to verify that a point we have reached is a candidate optimal solution.

- Given a point, it is easy to check which constraints are binding.

Fundamentals of optimization

Multivariate optimization-quadratic programming

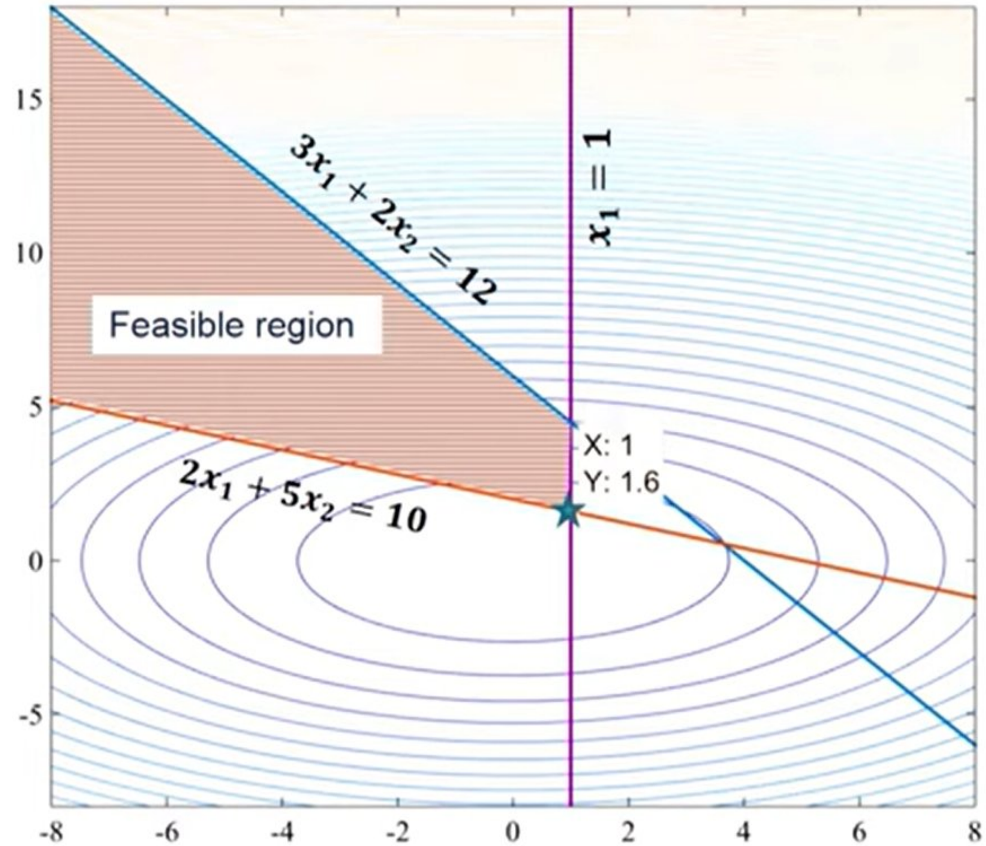
$$\min_{x_1, x_2} \; 2x_1^2 + 4x_2^2$$

$$st$$

$$3x_1 + 2x_2 \leq 12$$

$$2x_1 + 5x_2 \geq 10$$

$$x_1 \leq 1$$



Fundamentals of optimization

Multivariate optimization-quadratic programming

$$\min_{x_1, x_2} \quad 2x_1^2 + 4x_2^2$$

$$st$$

$$3x_1 + 2x_2 \le 12 \Rightarrow (a)$$

$$2x_1 + 5x_2 \ge 10 \Rightarrow (b)$$

$$x_1 \le 1 \Rightarrow (c)$$

- Lagrangian

$$L(x_1, x_2, \mu_1, \mu_2, \mu_3) = 2x_1^2 + 4x_2^2 + \mu_1(3x_1 + 2x_2 - 12) + \mu_2(10 - 2x_1 - 5x_2) + \mu_3(x_1 - 1)$$

- First order KKT conditions

$$4x_1 + 3\mu_1 - 2\mu_2 + \mu_3 = 0$$

$$8x_2 + 2\mu_1 - 5\mu_2 = 0$$

$$\mu_1(3x_1 + 2x_2 - 12) = 0$$

$$\mu_2(10 - 2x_1 - 5x_2) = 0$$

$$\mu_3(x_3 - 1) = 0$$

$$\mu_i \ge 0$$

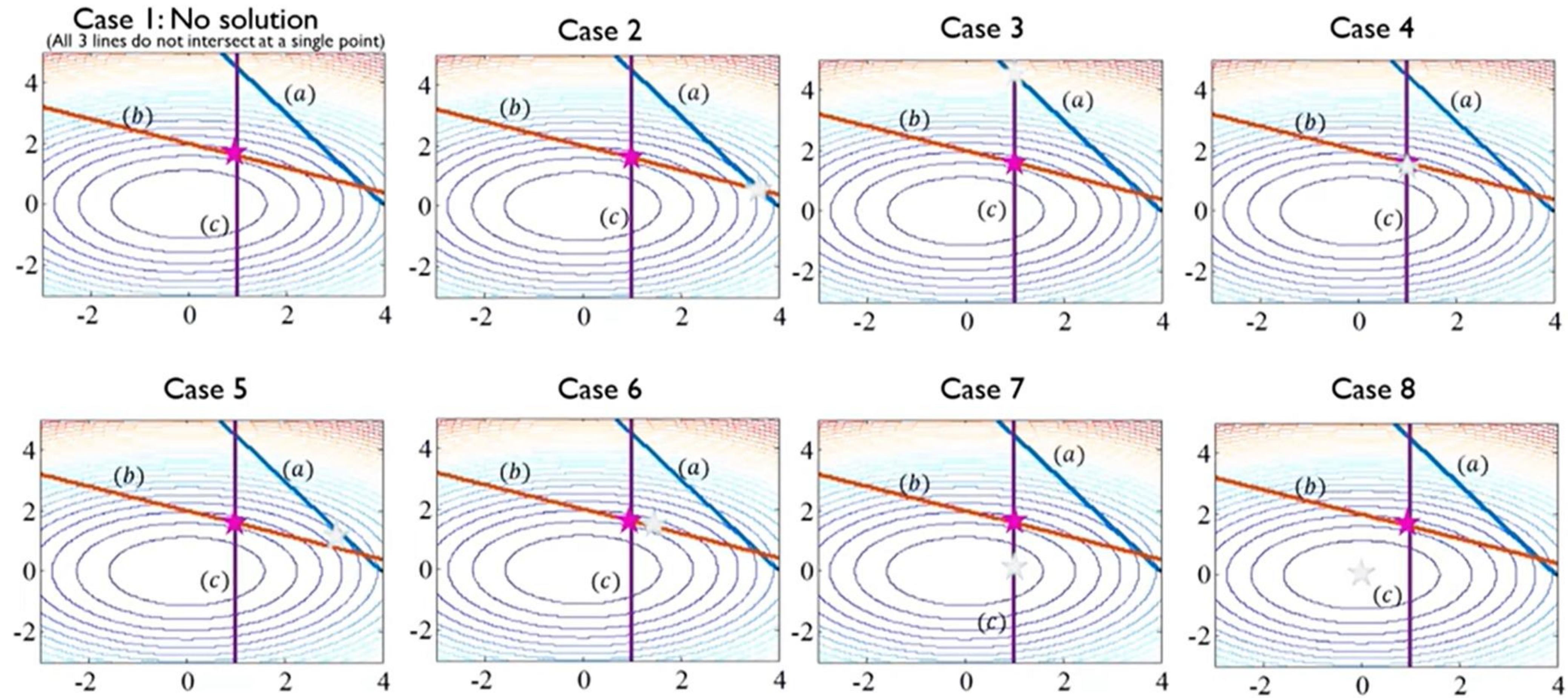
Fundamentals of optimization

Multivariate optimization-quadratic programming

Sl.no	Active (A) /Inactive (I) constraints (a)	(b)	(c)	Solution (x, μ)	Possible optima (Y/N)	Remark
1	A	A	A	Infeasible	N	Equations do not have a valid solution.
2	A	A	I	$x = [3.6364 \quad 0.5455]$ $\mu = [-5.2 \quad -1.45 \quad 0]$	N	$x_1 \leq 1$ is not satisfied, $\mu_1 < 0$, $\mu_2 < 0$
3	A	I	A	$x = [1 \quad 4.5]$ $\mu = [-18 \quad 0 \quad 50]$	N	$\mu_1 < 0$
4	I	A	A	$x = [1 \quad 1.6]$ $\mu = [0 \quad 2.56 \quad 1.12]$	Y	All constraints and KKT conditions satisfied
5	A	I	I	$x = [3.27 \quad 1.09]$ $\mu = [-4.36 \quad 0 \quad 0]$	N	$x_1 \leq 1$ is not satisfied
6	I	A	I	$x = [1.21 \quad 1.51]$ $\mu = [0 \quad 2.45 \quad 0]$	N	$x_1 \leq 1$ is not satisfied
7	I	I	A	$x = [1 \quad 0]$ $\mu = [0 \quad 0 \quad -4]$	N	$2x_1 + 5x_2 \geq 10$ is not satisfied
8	I	I	I	$x = [0 \quad 0]$ $\mu = [0 \quad 0 \quad 0]$	N	$2x_1 + 5x_2 \geq 10$ is not satisfied

Fundamentals of optimization

Solution for each case
Actual Optima



Introduction to Data Science

Techniques

- Regression analysis
- K-nearest-neighbor
- K-means clustering
- Logistics regression
- Principal Component Analysis
- Predictive Modeling
  - Lasso, Elastic net

Topics

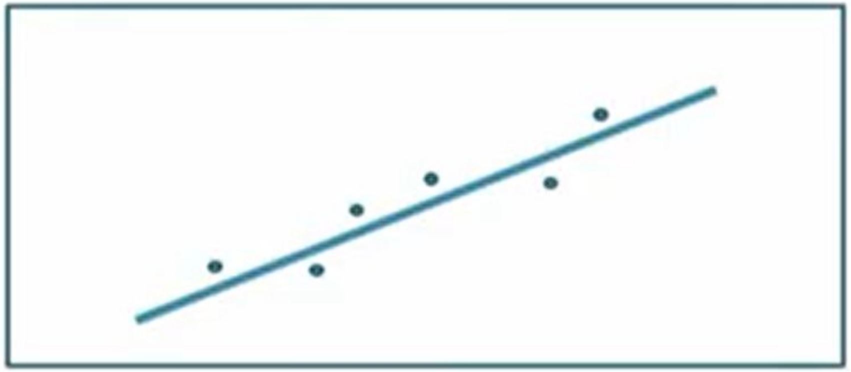
- Linear discriminant analysis (LDA)
- Support Vector Machines
- Decision trees and random forests
- Quadratic discriminant analysis (QDA)
- Naïve Bayes classifier
- Hierarchical clustering

What types of problems are being solved ? Why are there so many techniques?

Types of Problems

- Classification problems

- Function approximation



Thought Experiment

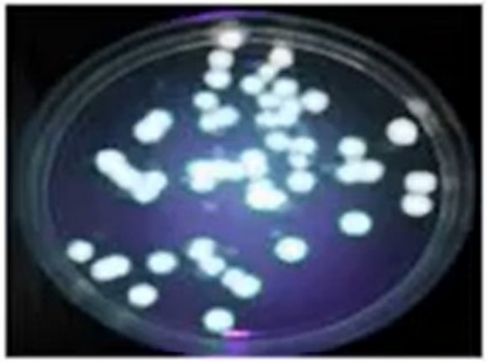
- How many articles are in the table?

- We can count all that is there to see

Thought Experiment (Metaphorical)

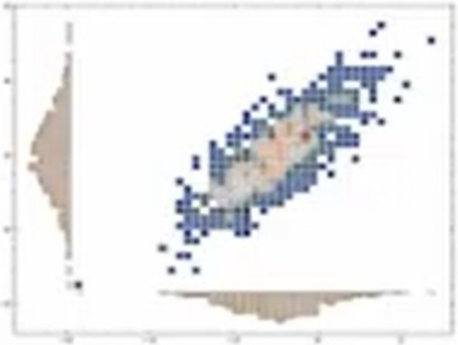
- What about things that we cannot see?

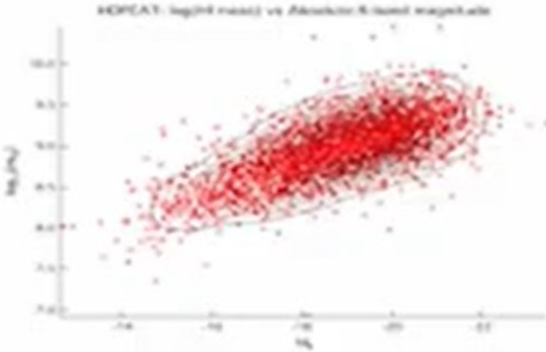
- How do we understand things that we cannot see – appropriate fluorescence chemical?

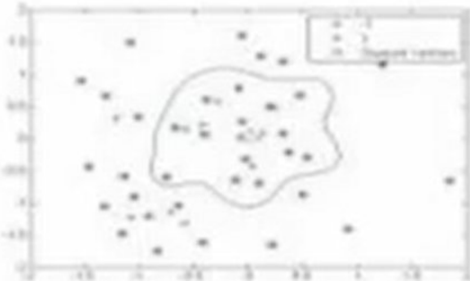


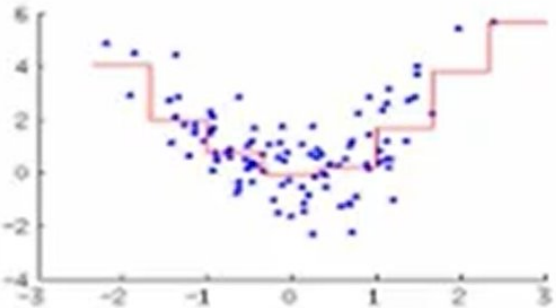
Thought Experiment

- If world were 2D?









- Data analytics not as critical

Thought Experiment

- Data analytics tools are like a microscope to probe higher dimensional data

- Make assumptions that has the possibility of characterizing the higher dimensional data
  - Gaussian distribution
  - Linearly separable
  - Many more

Thought Experiment

- Develop (Choose) a technique based on the assumptions that will satisfactorily answer questions about the data
- If the answers make sense then the data is "likely" to be organized in conformity with the assumptions
- If the answers do not make sense, modify assumptions and choose (develop) a technique
  - Hopefully, the previous iteration can be analyzed carefully in the assumption modification process
- Continue till the answers are satisfactory – Notice how we are seeing the "invisible"
- Understand the importance of test data in the process
- You now know why there are so many methods
  - Also tells you how you should choose a method

Solving Data Analysis Problems – A Guided Thought Process

Example - Data Imputation

- Readings from five sensors (X_1, X_2, X_3, X_4, X_5) are made available to you (for 100 different tests, check the file, *GTPvar.csv*). The readings are not arranged according to any order.

- There are some records, though, where there are a few missing readings that are marked *NA*.

- Your supervisor has asked you if there are any ideas that can be employed to rationally fill the missing values. Can you develop a data analytic approach to answer this question ?

Example - Data Imputation

- STEP 1: Problem Definition
  - Fill in missing data records
- STEP 2: Problem Characterization
  - Given part of the information, fill the missing information
  - Relate missing information with known information
  - Function approximation problem
  - $x_{\text{unknown}} = f\,(x_{\text{known}})$

Example - Data Imputation

- STEP 3: Solution Conceptualization
  - Need complete data set for identifying the function
    - Collect records without missing data
  - Assumption: All variables are independent of each other

    ⇒ no relation exists between the variables
  - For each variable, fill the missing data with the most likely value
- Step 3a: Verify assumption
  - Assumption not satisfied
- STEP 3: Solution Conceptualization
  - Assumption: Variables are inter-related
  - Step 3a: Assumption cannot be verified a priori

Example - Data Imputation

- STEP 4: Method Identification
  - Identify relationships using null space
  - Fill in missing values using the notion of pseudo-inverse
- STEP 5: Actualization
  - Implement in R programming language
- STEP 6 : Assess assumptions
  - Use it in intended application to check performance ?
- Solution realized (OR)
- STEP 3: .....
- STEP 4: …..
- STEP 5: ….
- STEP 6: ….

Conceptual Framework for Solving Data Analysis Problems

- START: Problem Arrival – Whole lot of words. Diffuse problem statement
- STEP 1: Problem Definition – Convert the loose words in to one problem statement (as precise as possible)
- STEP 2: Problem Characterization
  - Define high-level problems and sub-problems that need to be solved maintaining a high-level granularity
  - Develop a dependence diagram
  - Identify the problems and sub-problems as either function approximation or classification problems

Conceptual Framework for Solving Data Analysis Problems

- STEP 3: Solution Conceptualization – Visualization of the solution process through two conceptual devices
  - List assumptions (3a – Assumptions that can be verified a priori)
  - Flowchart
  - Pictures
- STEP 4: Method Identification – Map the elements of the flowchart and pictures into mathematical modules
  - Identify mathematical constructs/algorithms for the elements in the flowchart/picture
  - Identify lacunae – Data scientist to conceptualize method development
  - Develop the solution method map
- STEP 5: Actualization
  - Realize the solution method map in a software environment of choice
- STEP 6 : Assess assumptions and go through steps 3 to 6 if necessary

Conceptual Framework for Solving Data Analysis Problems